chromatography — separate ink colours

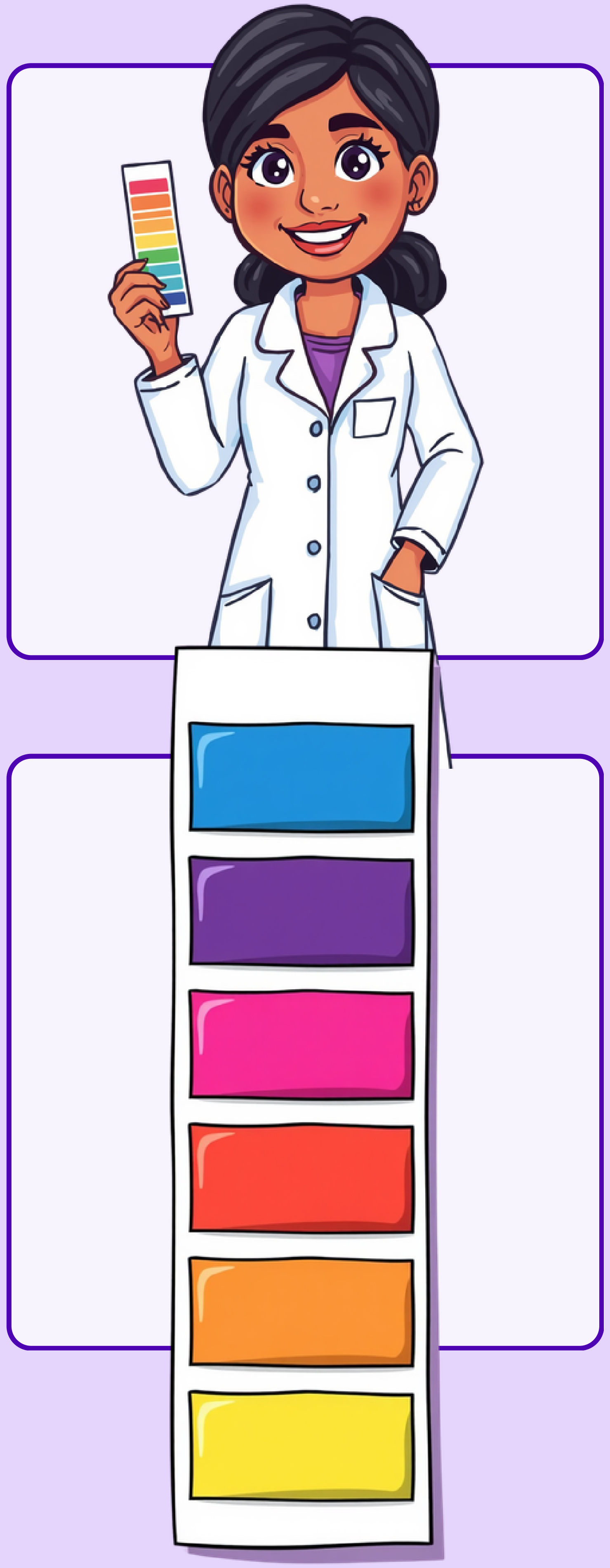
A Magical Science Experiment!

is black ink really black?

Let's discover the secret colours hiding inside!
Look at what happens when we put black ink on special paper and add water. The black colour breaks apart into a rainbow of hidden colours!
Are you ready to become a colour detective?

the big science word!

Chroma = Colour | Graphy = Writing
Chromatography means 'colour writing'!
A way scientists separate colours that are mixed together
It's like magic — finding hidden colours inside!



what do we need?

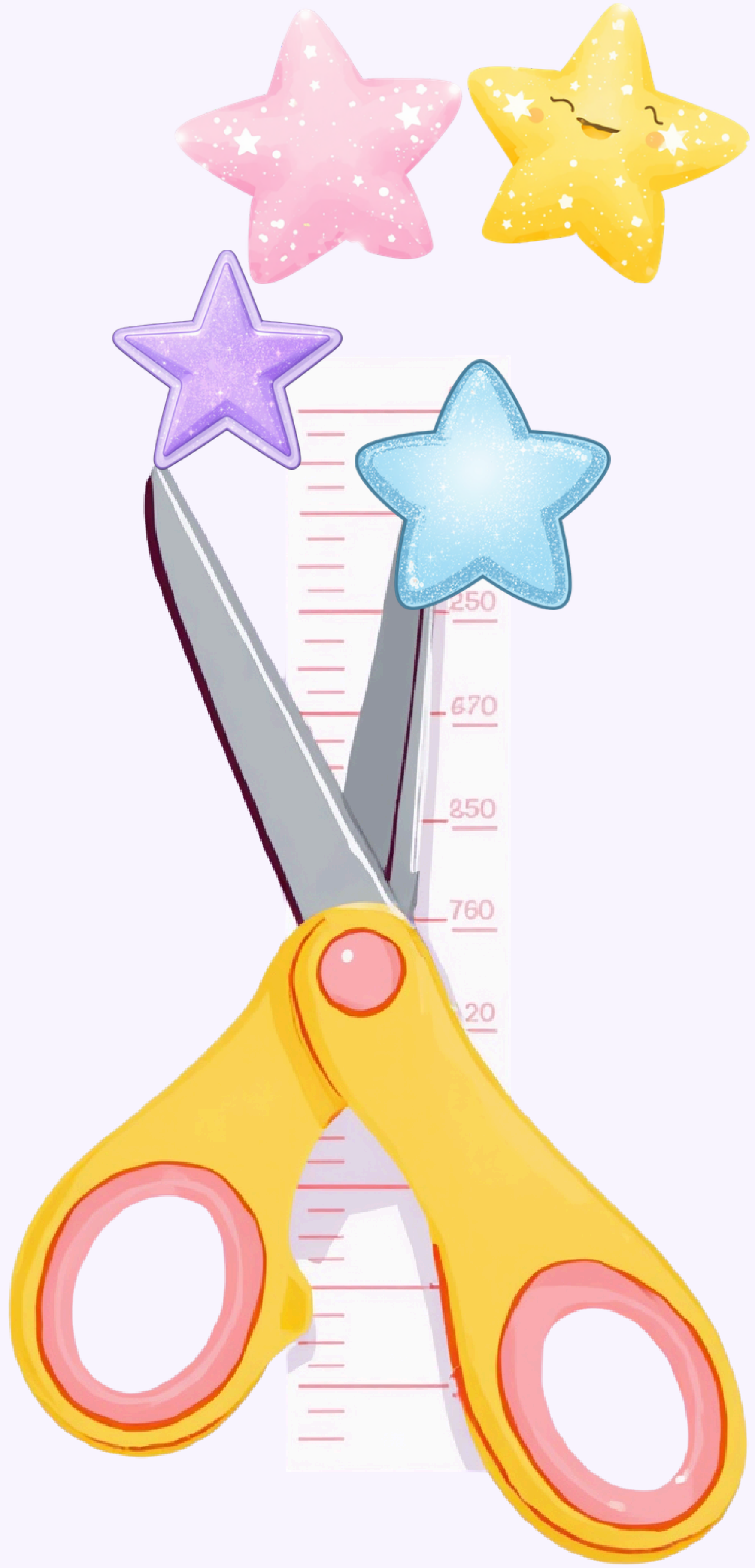
Black sketch pen (Camlin/Natraj/Reynolds)

Filter paper or coffee filter

Small bowl with water

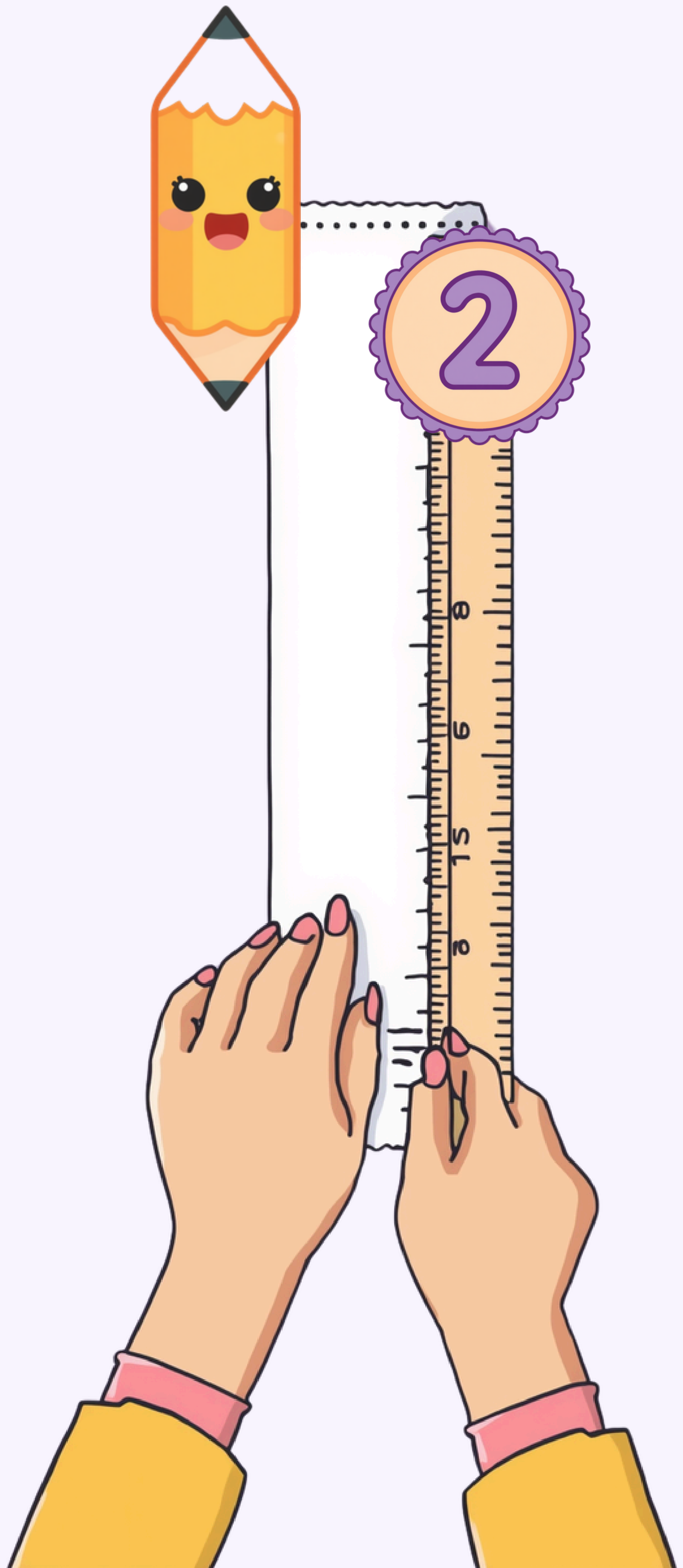
what do we need?

Pencil	Use a PENCIL to mark your paper — not a pen! Pencil lines won't mix with the water and spoil your experiment. A wooden pencil works best!
Ruler	Use a RULER to measure your paper strip and draw straight lines. This helps you be a careful scientist! Remember to measure 2 cm from the bottom.



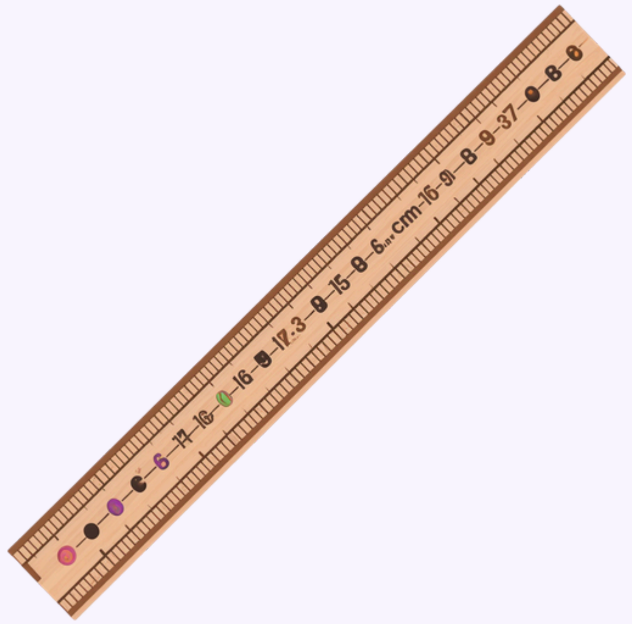
step 1 — prepare your paper

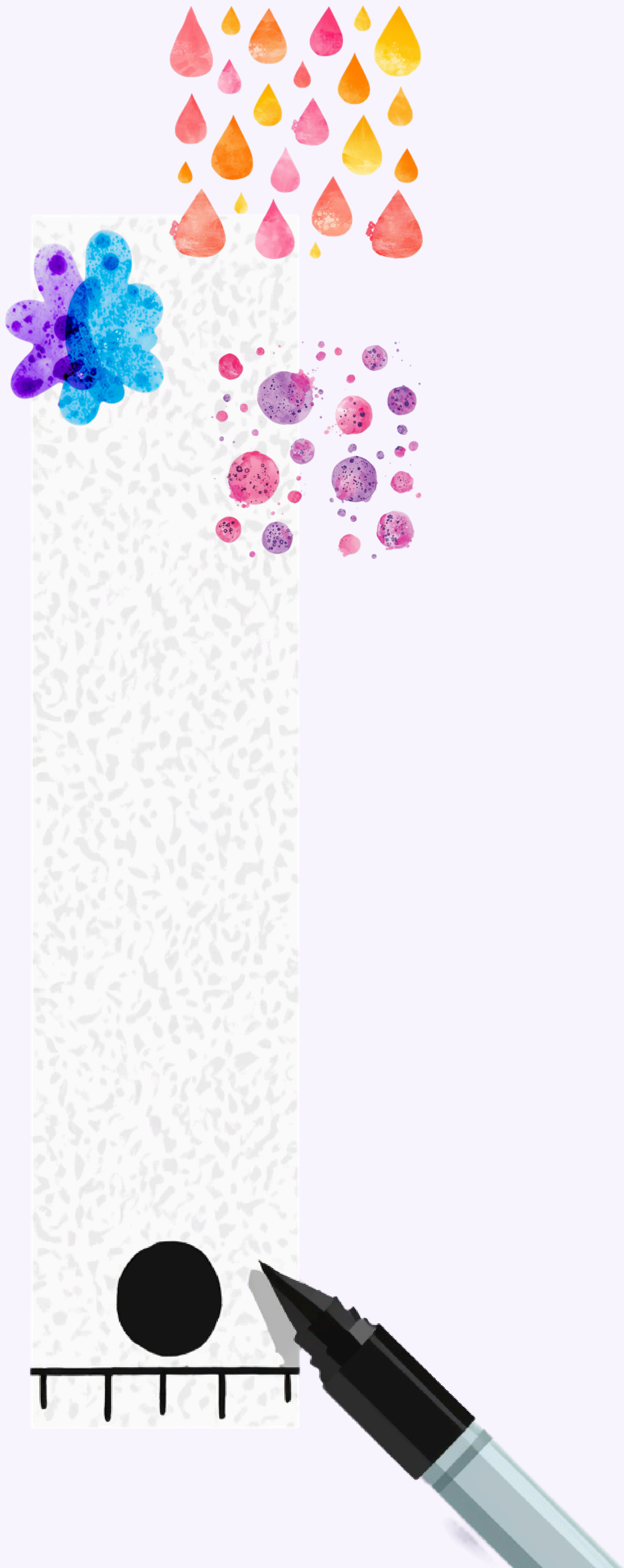
Cut your filter paper into a strip!
Make it 2 cm wide and 10 cm long.
Ask a grown-up to help you with the scissors. Measure carefully using your ruler!
This strip will be your magic colour paper!



step 2 — draw the start line

Use a PENCIL to draw a line 2 cm from the bottom of your paper strip.
Why pencil? Because pencil marks won't mix with water like ink does!
Place your ruler at the bottom of the paper strip and measure 2 cm up. Draw a straight line across the paper with your pencil.





step 3 — make the ink dot

Use your BLACK marker to make a thick dot right on the pencil line you drew!

Make the dot about 0.5 cm wide — that's about the size of a small pea.

Press firmly so the ink is nice and dark. This will help us see all the hidden colours better!

step 4 — add water

Important Tip! no. 01

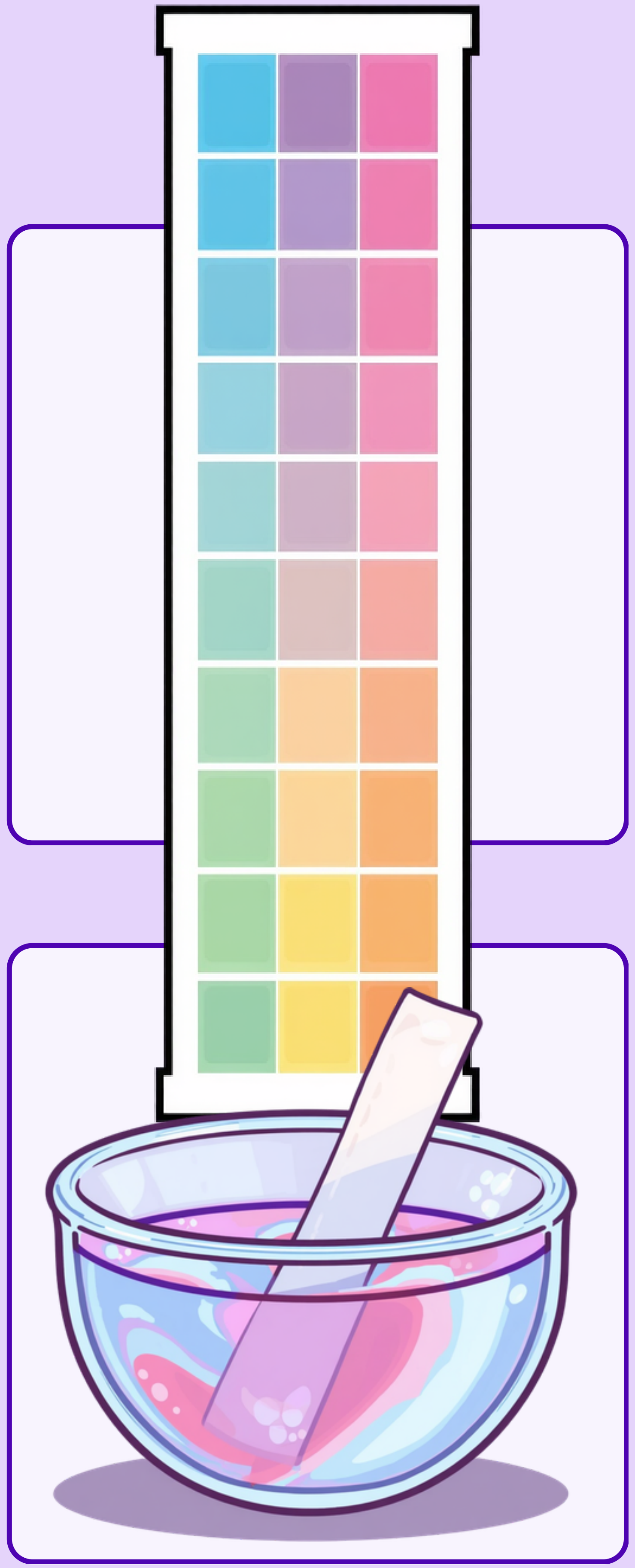
Pour water into (small bowl) — about 1 cm deep.
Very Important: The water level must be BELOW the ink dot! If the dot touches the water directly, the colours won't separate properly.

Important Tip! no. 02

Gently place your paper strip in the katori. The bottom of the paper should touch the water, but your ink dot should stay above the water line. Hold it steady!

step 5 — watch the magic!

Gently lower your paper strip into the water in the katori. Hold it steady!
Hold the paper strip upright and watch carefully. Don't let go!
The water climbs up the paper like magic! See it moving?
Wow! The colours start to separate! Watch the rainbow appear!



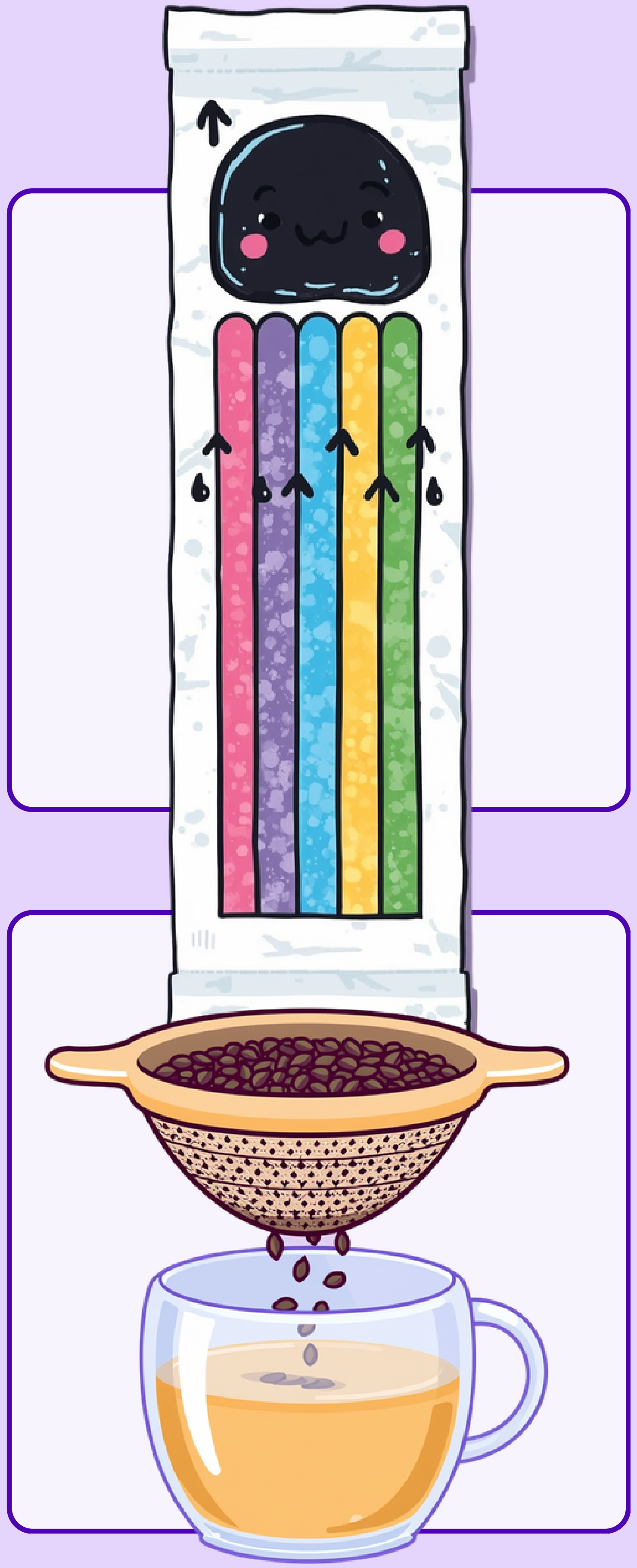
amazing colours appear!

How many colours did YOU see?

Blue	Purple	Orange
Yellow	Green	Red
Pink	Violet	More?

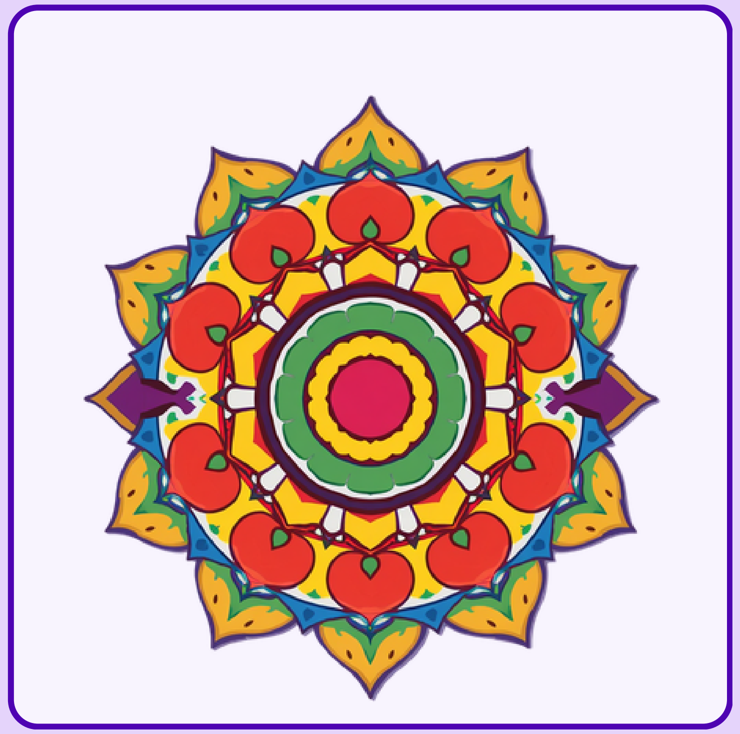
the science behind it!

Black ink is a MIXTURE of many colours hiding inside!
Water carries the ink up the paper like a tiny elevator.
Different colours travel at different speeds — some fast, some slow!
It's like separating chai leaves with a strainer — we separate the colours!



science in our daily life!

Rangoli colours mixed together — the beautiful patterns use many colours that blend and separate!
Holi gulal (coloured powder) — the bright colours we throw during Holi are mixtures too!
Scientists checking food colours in (sweets) — they make sure our favourite sweets are safe!
FSSAI scientists use chromatography to keep our food safe! They are colour detectives!



let's think like scientists!

What if we try a BLUE pen?

What if we use salt water?

What if we try different paper?

our science vocabulary!

Words we learned today!

CHROMATOGRAPHY	Separating colours	MIXTURE
Many things combined together	SEPARATION	Taking apart mixed things
OBSERVATION	Watching carefully like a detective	

you are scientists today!

Well Done ! Wah Wah! Super! Keep asking WHY and WHAT IF!